

Appendix

Appendix A: Details of Evaluation Metrics

A1. Manual Scoring Mechanism

To evaluate the quality of the generated results at different stages, this paper adopts a manual scoring mechanism to quantitatively assess the automated modeling results. The total score is set to 100 points and evenly divided into four components, with each component accounting for 25 points:

1. Natural-language modeling interpretation.
2. MG mathematical model generation.
3. Code conversion and solver execution.
4. Feedback correction and result completeness.

The scoring criteria for each component are defined as follows:

- 25 points are assigned if all requirements of the component are fully satisfied.
- 15 points are assigned if the component partially satisfies the requirements but contains minor errors, omissions, or irrelevant information.
- 0 points are assigned if the key requirements of the component are not satisfied.

Based on the above scoring rules, the comprehensive score of the i -th test can be expressed as:

$$S_i = S_i^{A_1} + S_i^{A_2} + S_i^{A_3} + S_i^L \quad (1)$$

where $S_i^{A_1}$, $S_i^{A_2}$, $S_i^{A_3}$, and S_i^L denote the scores of natural-language interpretation, mathematical model generation, code execution, and LangGraph-based feedback correction in the i -th test, respectively. The average score over all tests can be expressed as:

$$\bar{S} = \frac{1}{N} \sum_{i=1}^N S_i \quad (2)$$

where N denotes the total number of tests, and $\bar{S}$ represents the average score of the method across all test tasks.

A2. Pass@k Metric

In this paper, pass@1 represents the success rate of the first generation attempt, while pass@3 represents the success rate within three independent generations or feedback-revision attempts.

Considering the inherent stochasticity of LLM in automatically generating code and models, a single generation result is insufficient to fully reflect the potential capability of the method under multiple attempts. Therefore, this paper further adopts the pass@k metric to evaluate the probability that the proposed workflow obtains a valid result under multiple generations and revision attempts. For a given modeling task, if at least one of the k independent generation or revision attempts satisfies the predefined correctness criteria, the task is considered successful.

For a given task, suppose that n candidate results are generated in total, among which c candidates satisfy the above correctness criteria. Then, pass@k can be computed using the following unbiased estimator:

$$\text{Pass}@k = 1 - \frac{\binom{n-c}{k}}{\binom{n}{k}} \quad (3)$$

where k denotes the number of candidate results considered during evaluation. In the experimental setting of this paper, pass@1 represents the proportion of tasks in which a complete and executable microgrid modeling result can be obtained through a single generation attempt. pass@3 represents the proportion of tasks in which at least one complete and executable result is obtained within three independent generation or feedback-revision attempts.

Appendix B

B1: Agricultural MG Modeling

Consider the following natural-language modeling descriptions: Construct an agricultural MG model. The MG includes rooftop photovoltaic (PV), electrical storage system (ESS), biogas generator, boiler, and interactions with the external power grid. On the load side, the primary consideration is greenhouse agricultural production demands, including supplementary lighting, irrigation, ventilation, and heating. The optimization objective is to reduce the comprehensive operating cost of the agricultural MG while ensuring normal greenhouse production and considering low-carbon operation requirements. The model should account for electrical and thermal energy balance, storage and device operating limits, as well as greenhouse lighting and temperature requirements.

The *Task Interpreter Agent A_1* parses the above natural-language modeling requirement into a structured requirement specification. First, *Task Interpreter Agent A_1* identifies the current task as an agricultural microgrid modeling scenario. Then, it extracts the energy-side components from the text, including rooftop PV, ESS, biogas generator, boiler, and interactions with the external grid. It also identifies greenhouse agricultural loads such as supplementary lighting, irrigation, ventilation, and heating. The description “reduce comprehensive operating costs while considering low-carbon operation” is interpreted by *Task Interpreter Agent A_1* as an optimization objective that simultaneously minimizes operating costs and incorporates low-carbon operation considerations. Furthermore, electrical and thermal balance, energy network model, storage and device operational limits, as well as greenhouse lighting and temperature requirements, are identified as critical constraints that must be incorporated into subsequent mathematical modeling.

In this process, the knowledge graph provides *Task Interpreter Agent A_1* with standardized terminology and concept mapping support, enabling it to normalize expressions in natural language into a unified semantic form that can be directly utilized by subsequent modeling modules. For example, “boiler” is classified as a heating device, and “supplementary lighting, irrigation, ventilation, and heating” are identified as scenario-specific loads in the agricultural MG. Accordingly, *Task Interpreter Agent A_1* generates the following structured requirement specification:

< Type> Agricultural MG.

<Energy Devices> Rooftop PV, electric energy storage system(EES), biogas generator, and boiler

<Scenario-Specific Loads> Greenhouse supplementary lighting load, irrigation and fertigation load, ventilation load, greenhouse heating demand, monitoring and control load.

<Objective Functions> Minimize the comprehensive operating cost while considering low-carbon operation requirements.

<Constraints> Electrical power balance, thermal energy balance, operating boundaries of energy storage and other controllable devices, greenhouse lighting requirements, and greenhouse temperature requirements.

<Energy Network Model > Electric Network Model, Heating Network Model.

<Operating Mechanism and Coupling Relationships> The agricultural MG coordinates renewable generation, energy storage, biogas-based power generation, boiler heating, grid interaction, and greenhouse production loads. The electrical subsystem, thermal subsystem, and greenhouse operating demands are coupled through power balance, heat balance, and low-carbon operation requirements.

As a result, the *Task Interpreter Agent A₁* completes the transformation from the natural-language agricultural MG modeling requirement to the structured specification output Z_1 . This structured output explicitly defines the scenario type, device and load composition, optimization objectives, constraints, energy network model, and system coupling relationships, providing a direct basis for the subsequent *Mathematical Modeling Agent A₂*. In *Task Interpreter Agent A₁*, the knowledge graph performs terminology standardization, concept mapping, and scenario semantic organization, thereby improving the accuracy and consistency of the subsequent automated modeling workflow.

The *Task Interpreter Agent A₁* has already parsed the natural-language agricultural MG modeling requirement into the structured specification output Z_1 . The *Mathematical Modeling Agent A₂*, in conjunction with the scenario-oriented modeling knowledge graph, retrieves the relevant objective function components, device model templates, load model templates, and constraint knowledge for the agricultural MG and generates the MG mathematical model Z_2 .

a) Objective Function Formulation

Based on the structured specification output Z_1 , *Mathematical Modeling Agent A₂* generates the following objective function:

$$\min F_{\text{AMG}} = \sum_{t \in T} [C_{\text{GH}}(t) + C_{\text{DG}}(t) + C_{c,\text{utility}}(t) + C_{c,\text{biogas}}(t) - C_{c,\text{GH}}(t)] \quad (4)$$

where $C_{\text{GH}}(t)$ represents the cost associated with regulating greenhouse agricultural loads. $C_{\text{DG}}(t)$ denotes the operating cost of distributed energy resources. $C_{c,\text{utility}}(t)$, and $C_{c,\text{biogas}}(t)$ represent the cost of buying electricity from the grid and the costs related to carbon emissions resulting from biogas generation. $C_{c,\text{GH}}(t)$ represent the low-carbon benefits derived from greenhouse carbon absorption.

b) Device Model Formulation

Based on the device models required in structured specification output Z_1 , *Mathematical Modeling Agent A₂* retrieves the corresponding device model templates from the knowledge graph and generates the model formulations. For the biogas generator, its model formulation can be expressed as:

$$P_t^{\text{biogas}} = V_t^{\text{biogas}} \beta^{\text{biogas}} B C V^{\text{biogas}} \quad (5)$$

where P_t^{biogas} denotes the biogas generation power at time period t , V_t^{biogas} denotes the biogas flow rate, β^{biogas} denotes the biogas generation efficiency, and $B C V^{\text{biogas}}$ denotes the calorific value of biogas. For the electric boiler, the model can be expressed as:

$$Q_t^{\text{EB}} = \eta^{\text{EB}} P_t^{\text{EB}} \quad (6)$$

where Q_t^{EB} denotes the heat output of the electric boiler at time period t , P_t^{EB} denotes the electrical power consumed by the electric boiler, and η^{EB} denotes the electric-to-heat conversion efficiency. For the EES, the model can be expressed as:

$$E_t^{\text{EES}} = E_{t-1}^{\text{EES}} + \eta^{\text{ch}} P_t^{\text{ch}} \Delta t - \frac{P_t^{\text{dis}} \Delta t}{\eta^{\text{dis}}} \quad (7)$$

$$\underline{E}^{\text{EES}} \leq E_t^{\text{EES}} \leq \bar{E}^{\text{EES}} \quad (8)$$

$$0 \leq P_t^{\text{ch}} \leq \bar{P}^{\text{ch}}, \quad 0 \leq P_t^{\text{dis}} \leq \bar{P}^{\text{dis}} \quad (9)$$

where E_t^{EES} denotes the stored energy of the EES at time period t , P_t^{ch} and P_t^{dis} denote the charging and discharging power of the EES, respectively, and η^{ch} and η^{dis} denote the charging and discharging efficiencies, respectively. In addition, the operational boundaries of the devices are constructed as:

$$0 \leq P_t^{\text{PV}} \leq \bar{P}_t^{\text{PV}} \quad (10)$$

$$0 \leq P_t^{\text{biogas}} \leq \bar{P}^{\text{biogas}} \quad (11)$$

$$0 \leq Q_t^{\text{boiler}} \leq \bar{Q}^{\text{boiler}} \quad (12)$$

where P_t^{PV} denotes the utilized PV power at time period t , and $\bar{P}_t^{\text{PV}}$ denotes the upper limit of the available PV output. $\bar{P}_t^{\text{biogas}}$ represents the rated biogas generation power at time period t , and $\bar{Q}_t^{\text{boiler}}$ denotes the rated heat output of the boiler.

c) Load Model Formulation

The scenario-specific characteristics of agricultural MG are more prominently reflected on the load side, where greenhouse production loads exhibit clear environmental regulation properties and optimizable scheduling capabilities. Since rigid loads typically operate according to fixed demand profiles, they are not treated as the primary modeling objects. For flexible loads identified in structured specification output Z_I , such as supplementary lighting, irrigation, and ventilation, *Mathematical Modeling Agent A₂* retrieves the corresponding agricultural load models from the knowledge graph and constructs their mathematical expressions. First, the greenhouse electric load can be expressed as:

$$P_t^{\text{GH}} = P_t^{\text{light}} + P_t^{\text{irrigation}} + P_t^{\text{ventilation}} \quad (13)$$

where P_t^{GH} denotes the total electric load of the greenhouse, and P_t^{light} , $P_t^{\text{irrigation}}$, and $P_t^{\text{ventilation}}$ denote the supplementary lighting load, irrigation load, and ventilation load, respectively. The supplementary lighting load can be expressed as:

$$P_t^{\text{light}} = N_t^{\text{light}} P^{\text{lamp}} \quad (14)$$

where N_t^{light} denotes the number of supplementary lighting devices in operation at time period t , and P^{lamp} denotes the rated power of a single supplementary lighting device. The supplementary lighting load is not only a component of the electric load, but also directly affects the effective lighting level inside the greenhouse. Therefore, its model is further associated with the lighting constraint. The irrigation load can be expressed as:

$$P_t^{\text{irrigation}} = \frac{W_{h,t} \rho_w Z_{\text{pump}} g}{36 \eta_{\text{pump}}} \quad (15)$$

where $W_{h,t}$ denotes the irrigation water demand at time period t , ρ_w denotes the density of water, Z_{pump} denotes the pump head, g denotes the gravitational acceleration, and η_{pump} denotes the pump efficiency. The ventilation load can be expressed as:

$$P_t^{\text{ventilation}} = P^{\text{fan}} N_t^{\text{fan}} \quad (16)$$

where P^{fan} denotes the rated power of a single fan, and N_t^{fan} denotes the number of fans in operation at time period t . The ventilation load not only affects electric power consumption but also participates in the heat exchange process of the greenhouse, thereby further influencing the heating demand. For the greenhouse heating demand, the heat load model can be expressed as:

$$Q_t^{\text{GH}} = Q_t^{\text{enclosure}} + Q_t^{\text{ground}} + Q_t^{\text{ventilation}} - Q_t^{\text{lamp}} - Q_t^{\text{sun}} \quad (17)$$

where Q_t^{GH} denotes the total heating demand of the greenhouse; $Q_t^{\text{enclosure}}$, Q_t^{ground} , and $Q_t^{\text{ventilation}}$ denote the heat loss through the greenhouse enclosure, ground heat transfer loss, and ventilation heat loss, respectively; and Q_t^{lamp} and Q_t^{sun} denote the heat gains from supplementary lighting devices and solar radiation, respectively.

d) Constraint Construction

After constructing the energy-side device models and agricultural flexible load models, *Mathematical Modeling Agent A₂* further generates the system-level coupling relationships of the agricultural MG. The electrical power balance constraint can be expressed as:

$$P_t^{\text{PV}} + P_t^{\text{biogas}} + P_t^{\text{utility}} + P_t^{\text{dis}} = P_t^{\text{GH}} + P_t^{\text{ch}} + P_t^{\text{EB}} \quad (18)$$

where P_t^{utility} denotes the power purchased from the external grid by the agricultural MG. The thermal power balance constraint can be expressed as:

$$Q_t^{\text{EB}} = Q_t^{\text{GH}} \quad (19)$$

This equation indicates that the boiler heat output is used to satisfy the greenhouse heating demand. In addition, according to the greenhouse lighting and temperature requirements specified by *Task Interpreter Agent A₁*, the corresponding constraints can be further expressed as:

$$E_{a,\min} \leq E_{a,t} \leq E_{a,\max} \quad (20)$$

$$T_{\text{in},\min} \leq T_{\text{in},t} \leq T_{\text{in},\max} \quad (21)$$

where $E_{a,t}$ denotes the effective lighting intensity inside the greenhouse, and $T_{\text{in},t}$ denotes the indoor greenhouse temperature. These constraints are used to ensure that the greenhouse production environment satisfies the normal growth requirements of crops.

e) Electric and Heating Network Model

To further ensure the network-level feasibility of the generated agricultural MG model, *Mathematical Modeling Agent A₂* also constructs the internal electrical and heating network constraints. The network constraints describe the power/heat transfer process inside the MG, including power flow and thermal flow. For the internal electrical network of the agricultural MG, a simplified power-flow model can be expressed as:

$$P_{ij,t}^e = \sum_{k:j \rightarrow k} P_{jk,t}^e + P_{j,t}^{\text{load}} - P_{j,t}^{\text{gen}} \quad (22)$$

$$Q_{ij,t}^e = \sum_{k:j \rightarrow k} Q_{jk,t}^e + Q_{j,t}^{\text{load}} - Q_{j,t}^{\text{gen}} \quad (23)$$

$$V_{j,t} = V_{i,t} - 2(r_{ij} P_{ij,t}^e + x_{ij} Q_{ij,t}^e) \quad (24)$$

$$V_i^{min} \leq V_{i,t} \leq V_i^{max} \quad (25)$$

where $P_{ij,t}^e$ and $Q_{ij,t}^e$ denote the active and reactive power flows from node i to node j , respectively; $V_{i,t}$ denotes the voltage magnitude at node i ; and r_{ij} and x_{ij} denote the resistance and reactance of branch ij , respectively. For the internal heating network, the heat delivered to node n can be described by the mass flow rate and the temperature difference between the supply and return pipelines:

$$Q_{n,t}^h = c_w m_{n,t} (T_{n,t}^s - T_{n,t}^r) \quad (26)$$

The heat loss along pipeline l can be represented as:

$$T_{l,t}^{out} = T_t^{amb} + (T_{l,t}^{in} - T_t^{amb}) \exp\left(-\frac{\lambda_l L_l}{c_w m_{l,t}}\right) \quad (27)$$

$$T_n^{s,min} \leq T_{n,t}^s \leq T_n^{s,max} \quad (28)$$

$$T_n^{r,min} \leq T_{n,t}^r \leq T_n^{r,max} \quad (29)$$

where $Q_{n,t}^h$ denotes the heat delivered to node n , c_w is the specific heat capacity of water, $m_{n,t}$ denotes the mass flow rate, $T_{n,t}^s$ and $T_{n,t}^r$ denote the supply and return temperatures, respectively. $T_{l,t}^{in}$ and $T_{l,t}^{out}$ denote the inlet and outlet temperatures of pipeline l , T_t^{amb} is the ambient temperature, λ_l is the heat loss coefficient, and L_l is the pipeline length. In this way, *Mathematical Modeling Agent A₂* generates the mathematical optimization model MG mathematical model Z_2 for the agricultural MG scenario.

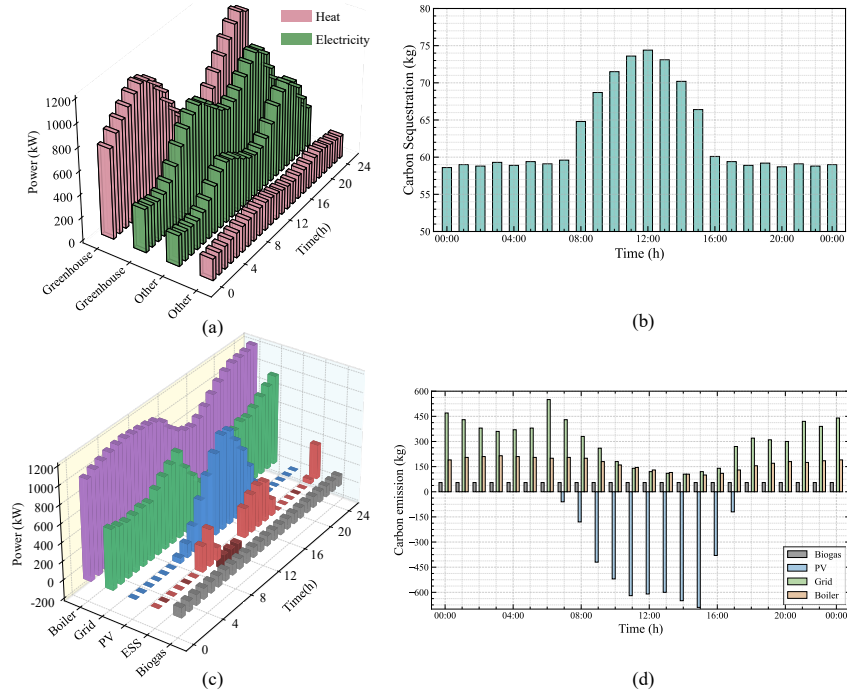


Fig. B1. (a) Rigid load and flexible load power distribution, (b) Carbon emission distribution of the greenhouse, (c) Electrical power scheduling of the agricultural MG, (d) Carbon emission of distributed energy resources.

B2: Evaluation Metrics for Knowledge Coverage and Utilization

To quantitatively evaluate the quality of the generated models, three metrics are adopted, including modeling knowledge coverage, constraint knowledge coverage, and knowledge utilization efficiency. For the i -th scenario, let $\mathcal{M}_i^{\text{ref}}$ and $\mathcal{M}_i^{\text{gen}}$ denote the reference modeling knowledge set and the generated modeling knowledge set, respectively. The modeling knowledge coverage is defined as:

$$\text{MKC}_i = \frac{|\mathcal{M}_i^{\text{ref}} \cap \mathcal{M}_i^{\text{gen}}|}{|\mathcal{M}_i^{\text{ref}}|} \quad (30)$$

Similarly, let $\mathcal{C}_i^{\text{ref}}$ and $\mathcal{C}_i^{\text{gen}}$ denote the reference constraint knowledge set and the generated constraint knowledge set, respectively. The constraint knowledge coverage is calculated as:

$$\text{CKC}_i = \frac{|\mathcal{C}_i^{\text{ref}} \cap \mathcal{C}_i^{\text{gen}}|}{|\mathcal{C}_i^{\text{ref}}|} \quad (31)$$

In addition, let $\mathcal{K}_i^{\text{ret}}$ be the retrieved knowledge set and $\mathcal{K}_i^{\text{use}}$ be the subset of retrieved knowledge that is effectively used in the generated model. The knowledge utilization efficiency is defined as:

$$\text{KUE}_i = \frac{|\mathcal{K}_i^{\text{use}}|}{|\mathcal{K}_i^{\text{ret}}|} \quad (32)$$

For all N testing scenarios, the final reported values are obtained by averaging the corresponding scenario-level metrics. Thus, the overall evaluation metrics are given by:

$$\text{MKC} = \frac{1}{N} \sum_{i=1}^N \text{MKC}_i, \quad \text{CKC} = \frac{1}{N} \sum_{i=1}^N \text{CKC}_i, \quad \text{KUE} = \frac{1}{N} \sum_{i=1}^N \text{KUE}_i \quad (33)$$

B3: Multi-Energy MG Modeling

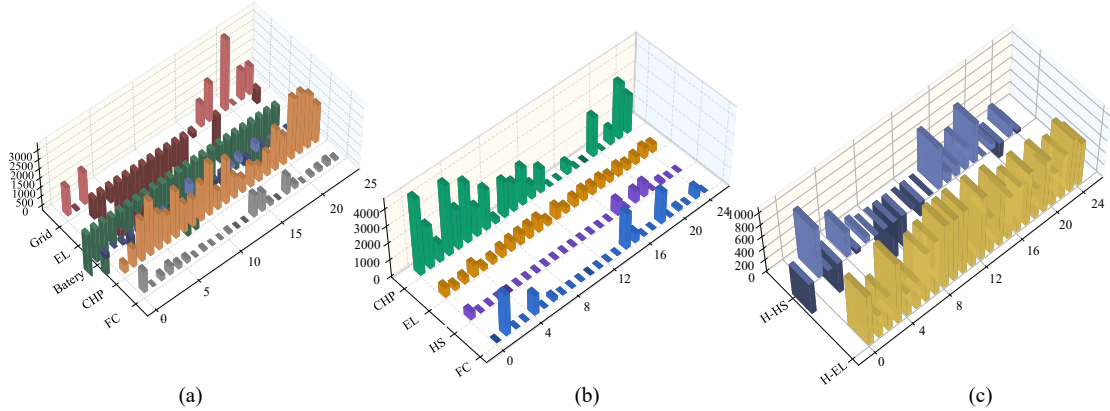


Fig. B2. (a) Electrical power scheduling of the multi-energy MG, (b) Thermal scheduling of the multi-energy MG, (c) Hydrogen scheduling of the multi-energy MG.